

Model	Microbiome	Metabolomics	Proteomics
LR age-only	0.691 (0.608–0.775)	0.577 (0.453–0.701)	0.573 (0.453–0.694)
RF clinical	0.775 (0.700–0.849)	0.709 (0.602–0.816)	0.647 (0.533–0.762)
EN omics	0.620 (0.532–0.708)	0.722 (0.617–0.828)	0.695 (0.584–0.805)
XGB omics	0.586 (0.495–0.678)	0.737 (0.633–0.840)	0.674 (0.563–0.786)
RF clinical + EN	0.769 (0.694–0.845)	0.750 (0.650–0.851)	0.703 (0.595–0.811)
RF clinical + XGB	0.755 (0.677–0.832)	0.770 (0.670–0.870)	0.705 (0.597–0.812)

AUC with 95% DeLong confidence interval on the held-out test set. LR age-only = logistic regression on age only.